

Post-Concussion Brain Changes Relative to Pre-Injury White Matter and Cerebral Blood Flow

A Prospective Observational Study

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Abstract

Background and Objectives

Medical clearance for return to play (RTP) after sports-related concussion is based on clinical assessment. It is unknown whether brain physiology has entirely returned to preinjury baseline at the time of clearance. In this longitudinal study, we assessed whether concussed individuals show functional and structural MRI brain changes relative to preinjury levels that persist beyond medical clearance. Secondary objectives were to test whether postconcussion changes exceed uninjured brain variability and to correlate MRI findings with clinical recovery time.

Methods

For this prospective observational study, healthy athletes without a history of psychiatric, neurologic, or sensory-motor conditions were recruited from a single university sport medicine clinic. Clinical and MRI data were collected at preseason baseline, and those who were later concussed were reassessed at 1–7 days after injury, RTP, 1–3 months after RTP, and 1 year after RTP. A demographically matched control cohort of uninjured athletes was also reassessed at their subsequent preseason baseline. Primary outcomes were postconcussion changes in MRI measures of cerebral blood flow (CBF), white matter mean diffusivity (MD), and fractional anisotropy (FA), evaluated using mixed models. Secondary outcomes were group differences in MRI change scores and correlations of change scores with days to RTP.

Results

Of the 187 athletes enrolled in the study, 25 had concussion with follow-up imaging (20.3 ± 1.5 years, 56% male, 44% female) and were compared with 27 controls (19.7 ± 1.8 years, 44% male, 56% female). Concussed athletes showed statistically significant changes from baseline, including decreased frontoinsular CBF (mean and 95% CI $-8.97 [-12.80, -5.01]$ mL/100 g/minute, $z = -4.53$), along with increased MD ($1.94 \times 10^{-5} [1.26, 2.69] \times 10^{-5}$, $z = 5.48$) and reduced FA ($-7.30 \times 10^{-3} [-9.80, -5.05] \times 10^{-3}$, $z = -6.07$) in the corona radiata and internal capsule. Effects persisted beyond RTP, although only CBF changes exceeded longitudinal variability in controls. For participants with longer recovery periods, significantly greater changes in medial temporal CBF were also seen ($\rho = 0.64 [0.44, 0.81]$, $z = 6.80$).

Discussion

This study provides direct evidence of persistent postconcussion changes in CBF and white matter at RTP and up to 1 year later. These results support incomplete recovery of brain physiology at medical clearance, with secondary analyses emphasizing the sensitivity of CBF to clinical recovery.

RELATED ARTICLE

Editorial

Uncovering Subclinical Neural Alterations in Sport-Related Concussion: The Added Value of Longitudinal Neuroimaging

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Glossary

ACU = acute injury; ASL = arterial spin labeling; BSR = bootstrap ratio; CBF = cerebral blood flow; CoM = center of mass; CRT = choice reaction time; DTI = diffusion tensor imaging; FA = fractional anisotropy; FDR = false discovery rate; HOC = history of concussion; LMM = linear mixed modeling; M-BESS = Modified Balance Error Scoring System; MD = mean diffusivity; MNI = Montreal Neurological Institute; MPRAGE = magnetization-prepared rapid acquisition gradient echo imaging; PS = processing speed; RTP = return to play; SAC = Standardized Assessment of Concussion; SCAT = Sport Concussion Assessment Tool; SYM = symptomatic injury; TMTs A and B = Trail Making Tests A and B.

Introduction

Concussion is a major health concern, characterized by acute disturbances but also long-term effects on brain function. At present, clinical determination of postconcussion recovery is based on symptom assessments, combined with brief tests of cognition, balance, and visual function.¹ There is growing recognition, however, that physiologic brain recovery may persist beyond the resolution of clinical symptoms, with MRI studies showing disturbances months to years after clinical resolution.^{1,2} This raises concerns about possible long-term effects of concussion on neural physiology, the consequences of which may be exacerbated by repetitive injury.^{3,4}

At present, most neuroimaging evidence for persistent postconcussion brain changes is based on cross-sectional studies comparing concussed individuals with uninjured controls. This approach, however, is confounded by baseline differences in individual brain physiology. This includes demographic effects of age, sex, and concussion history,⁵⁻⁷ but also individual-level differences in neurophysiology.⁸ To control these issues, prospective analyses are needed that compare baseline and postconcussion imaging within the same individuals; but, such studies are challenging and remain rare in the literature. A series of studies examined a cohort of 45 hockey players, of whom 11 were concussed and followed from 72 hours to 2 months after injury, with persistent alterations in white matter microstructure, neurometabolites, and gray matter volume identified.⁹⁻¹² To date, however, brain changes remain under-examined relative to the clinical determination of recovery. Moreover, there is a need for more prospective studies that incorporate multiple MRI measures for a more complete picture of this complex neurologic condition.¹³

This study addressed this gap, using MRI to image a large cohort of varsity athletes at the beginning of their athletic season; those who were later concussed were reimaged, beginning at early injury and following them up to 1 year after medical clearance to return to play (RTP). Imaging included arterial spin labeling (ASL) to quantify resting cerebral blood flow (CBF) and diffusion tensor imaging (DTI) to assess white matter microstructural indices of mean diffusivity (MD) and fractional anisotropy (FA). While many MRI measures of brain physiology have shown persistent post-RTP effects,² ASL and DTI are of particular interest because perfusion

deficits and microstructural white matter changes are considered key mechanistic contributors to acute injury (ACU)¹³ and long-term neuropathologic change.^{14,15} This is supported by a substantial body of neuroimaging research,^{16,17} showing both acute and long-term postconcussion alterations.

From these parameters, we tested the principal hypothesis that (1) concussed individuals show altered CBF, MD, and FA relative to preinjury levels, with effects that persist beyond medical clearance. Secondary hypotheses were that postconcussion brain changes (2) exceed longitudinal brain changes among uninjured controls and (3) correlate with time to medical clearance (i.e., reflecting the severity of clinical outcome).

Methods

Study Participants

A cohort of university-level athletes was enrolled in this study, with recruitment and data collection spanning October 2014 to November 2019. Baseline recruitment aimed for 170–200 participants, given a target of 20–24 concussed athletes, with target numbers based on previous cross-sectional work.¹⁸ Athletes were recruited consecutively from multiple sport teams at the University of Toronto through the institutional sport medicine clinic (eTable 1 provides numbers by sport). Participants were approached during mandatory preseason baseline testing at the clinic and enrolled in the study provided that they met eligibility criteria, which included no history of neurologic disorders, psychiatric disease, sensory/motor impairments, or MRI contraindications. Baseline neuroimaging was conducted at this time, and clinical data were collected, including tests of symptoms, balance, and cognition.

Study participants were monitored by clinic staff over the course of their varsity career. Athletes who were later concussed were approached for reimaging, after diagnosis by a staff physician trained to assess sport-related concussion and confirmation that they did not exhibit gross postconcussion neurologic impairments. Follow-up imaging was conducted at early symptomatic injury (SYM), medical clearance to RTP, 1–3 months after RTP (1-3MO), and 1 year after RTP (1 YR). Early symptomatic imaging took place a median of 5 days after the concussion event (interquartile range [3, 6] days). Clinical data were also collected at SYM and RTP. Among athletes without documented concussion, a subset

was recontacted by study personnel at the time of subsequent preseason baseline testing, with follow-up neuroimaging and clinical data collection. While the concussed group included all eligible individuals with concussion after baseline scanning, the control group was selected to match on sample size and demographics. The baseline-to-baseline interval for controls was chosen to mitigate effects of within-season training load, which may substantially alter neurophysiology.¹⁹ The absence of concussion between control scans was confirmed by coaches, clinic staff, and spotters at practices and competitions, all of whom were trained to identify concussion signs and symptoms.

Standard Protocol Approvals, Registrations, and Patient Consents

The study was conducted in accordance with Canadian Tri-Council Policy Statement 2 and with approval of the University of Toronto and St. Michael's Hospital research ethics boards (REB# 17–228), with all study participants giving free and written informed consent. This study was written in accordance with Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.²⁰

MRI

Neuroimaging was conducted on a 3T MRI system (Magnetom Skyra, Siemens Healthineers) with a 20-channel head coil. Acquisition and preprocessing details are provided in eAppendix 1. Anatomical imaging included T1-weighted magnetization-prepared rapid acquisition gradient echo imaging ([MPRAGE]; sagittal acquisition, 0.9×0.9 mm in-plane, 0.9-mm slice thickness), fluid-attenuated inversion recovery imaging (FLAIR; sagittal acquisition, 0.4×0.4 mm in-plane, 0.9-mm slice thickness), and susceptibility-weighted imaging (SWI; oblique-axial acquisition, 0.6×0.6 mm in-plane, 1.2-mm slice thickness) sequences. Structural images were inspected by an MRI technologist and later reviewed by a neuroradiologist, with clinical reporting if abnormalities were identified. No abnormalities (white matter hyperintensities, contusions, microhemorrhage) were reported.

Resting blood flow was evaluated by 2D multislice pulsed ASL ($4 \times 4 \times 8$ mm voxels, 45 tag-control pairs, $TI1/TI2 = 700/1800$ ms), producing maps of absolute CBF in mL/100 g/minute. The maps were warped into Montreal Neurological Institute (MNI) coordinate space and voxels resampled to 3-mm isotropic resolution. Afterward, the CBF maps were parcellated using the Brainnetome Atlas, which subdivides cerebral gray matter into 246 cortical and subcortical regions.²¹ The mean CBF value was then computed for voxels in each parcel of the atlas for further analysis, reducing the number of statistical tests and improving local detection power. For 1 concussed study participant, MPRAGE data were not collected at baseline. To process these data, a modified FLAIR-based warping pipeline was used (details in eAppendix 1). White matter was evaluated using diffusion MRI ($2 \times 2 \times 2$ mm voxels, 30 diffusion-encoding directions, $b = 700$ seconds/mm²), producing maps of microstructural

indices including MD and FA. These maps were warped into MNI coordinate space and voxels resampled to 2-mm isotropic resolution. Afterward, the MD and FA maps were parcellated using the John Hopkins University ICBM-DTI-81 white matter atlas,²² which subdivides cerebral white matter into 48 regions, and the mean MD and FA values were computed for voxels within each parcel.

Data Checks

Two rounds of outlier testing were conducted to identify (a) scans showing evidence of extreme head motion and related signal effects and (b) poor fit of perfusion and diffusion models, reflected in extreme CBF, MD, or FA values (eAppendix 1 describes testing details). For CBF data, two outliers were identified (control scans, at follow-up), and for MD data, one outlier was identified (control scan, at follow-up) while no FA outliers were identified. The outlier scans were excluded from further analysis of concussion effects. Further post hoc testing of head motion found no significant confounding effects for the main study analyses (eAppendix 1 describes details).

Demographics and Clinical Data

For all participants, demographic information was collected at baseline, including age, sex, sport type, history of concussion (HOC), number of prior concussions, time since last injury, and recovery time from last injury. Clinical testing was conducted using C3 Logix, an iPad-based application that assesses symptoms (Sport Concussion Assessment Tool [SCAT] evaluation scale), cognition (Standardized Assessment of Concussion [SAC]), and balance (Modified Balance Error Scoring System [M-BESS]), with data collected at preseason baseline for all athletes, along with ACU and RTP for concussed athletes. Per evolving clinical guidelines, athletes were assessed with the most recent SCAT version at the time of enrollment (SCAT3 or SCAT5). For symptoms and balance, analyses used all data, focusing on the 22-item symptom list common to both test versions. Analysis of cognitive scores was restricted to SCAT5 data, which used 10-word lists. C3 Logix also includes computerized neurocognitive testing,²³ including simple reaction time (SRT) and choice reaction time (CRT), processing speed (PS), and Trail Making Test A and B (TMT A and B) completion times.

Analysis of Demographic and Clinical Data

Demographic data were summarized for concussed and control cohorts and, for the concussed athletes, missing scans were reported. Baseline demographics, HOC characteristics, and clinical scores were compared between groups using nonparametric 2-sample Wilcoxon tests. For clinical measures, the SCAT domain scores (number of symptoms, symptom severity, cognitive score, balance errors) of concussed athletes were compared with their own preinjury baseline, at ACU and at RTP, using paired-measures Wilcoxon tests. Similar testing compared C3 Logix neurocognitive tests of SRT and CRT, PS, and TMT A and B completion times, at ACU and RTP, relative to baseline. For all tests, *p* values and approximate *z*-scores were reported

and significance was determined after adjusting for multiple comparisons at a false discovery rate (FDR) of 0.05.

Analysis of Neuroimaging Data

To model postconcussion changes in CBF, FA, and MD relative to preinjury baseline, the effects of imaging sessions on MRI parameters were estimated through linear mixed modeling (LMM), which handles missing data by maximum likelihood estimation. The model included fixed-effect changes in MRI values at postconcussion imaging sessions (SYM, RTP, 1-3MO, 1 YR) relative to preinjury baseline, along with random-effect subject intercepts. The model is coded as

$$y \sim 1 + \text{SYM} + \text{RTP} + 1\text{-}3\text{MO} + 1\text{YR} + (1|\text{participant})$$

and obtains fixed-effect coefficients b_{SYM} , b_{RTP} , $b_{1\text{-}3\text{MO}}$, $b_{1\text{YR}}$. The models were fitted using the Matlab's R2022a *fitlme* package (The MathWorks, Natick, MA) with bootstrap resampling, where a resampling unit consisted of all sessions for a participant. This produced empirical distributions of fixed-effect estimates b , with corresponding p values. For each session, brain regions showing significant longitudinal change were identified at FDR = 0.05 and regional effect sizes reported as bootstrap ratio (BSR) values (a z-score statistic of effect).

For each postconcussion session, effect sizes were also summarized by averaging MRI parameter values over significant brain regions; refitting a bootstrapped linear mixed model on average values; and then reporting a summary b value, 95% CI, BSR, and p value for the appropriate postconcussion session (denoted $b_{\text{SYM,av}}$, $b_{\text{RTP,av}}$, $b_{1\text{-}3\text{MO,av}}$, $b_{1\text{YR,av}}$). For comparison, effect sizes in the control group, which had 2 beginning-of-season scans (BOS1, BOS2), were similarly summarized. After averaging connectivity values over significant connections and fitting a bootstrapped linear mixed model on $y \sim \text{BOS2} + (1|\text{participant})$, summary statistics were obtained for the fixed-effect coefficient of BOS2 (denoted $b_{\text{CTL,av}}$). The effect of adding model covariates adjusting for age, sex, HOC, and collision sport participation was assessed on these summary models, by calculating the difference in session-effect coefficients b , for linear mixed models with/without the covariates included; significance was evaluated by obtaining bootstrapped p values for these differences. None of the session effects was significantly altered at $p < 0.05$ (uncorrected), and these covariates were omitted from the final models.

To further assess whether postconcussion changes in CBF, MD, and FA exceed uninjured brain variability, cross-sectional analyses compared the longitudinal change scores of concussed athletes and controls, focusing on brain regions altered at RTP. MRI parameter values were averaged over regions with significant b_{RTP} values based on previous LMM analyses. The average values were calculated for every participant and imaging session, and then, group differences in mean change scores were calculated (denoted $\Delta_{\text{SYM-CTL}}$, $\Delta_{\text{RTP-CTL}}$,

$\Delta_{1\text{-}3\text{MO-CTL}}$, $\Delta_{1\text{YR-CTL}}$) with bootstrapped 95% CIs in 2-sample tests. Missing participant data were accounted for using bootstrapped multiple imputation, with linear mixed models fitted on average parameter values to simulate MRI data. The impact of adding covariates of age, sex, HOC, and collision sport participation to the imputation models was assessed as previously described. None of the between-group effects was significantly altered at $p < 0.05$ (uncorrected), and these covariates were omitted from the final models.

Finally, characteristics of the control group were further evaluated by whole-brain analyses. Paired-measures bootstrap analyses assessed longitudinal change from BOS1 to BOS2, and two-sample bootstrap analyses assessed group differences in baseline values between concussed and control groups, with significance determined at FDR = 0.05. Detailed reporting of significant ROIs for all analyses is provided in eAppendix 2.

Analysis of Neuroimaging and Recovery Time

Given the substantial variability at RTP, secondary analyses tested for associations between postconcussion changes in MRI parameter values and recovery time. This was examined for the number of days to RTP, transformed into base-10 log-scale to stabilize parameter estimates in the presence of heavy distribution tails because the values were initially found to deviate substantially from normality (skew = 1.8, kurtosis = 5.2), which was corrected by the transform (skew = -0.2, kurtosis = 2.7). Pearson correlations were calculated between log-dRTP and the MRI parameter change scores at each postconcussion imaging session, with bootstrap resampling to obtain empirical p values, and with significant regions identified at FDR = 0.05. Afterward, overall effects were reported for each session, by averaging MRI values over significant brain regions and refitting a bootstrapped correlation model, producing a summary correlation value, 95% CI, p value, and BSR. Exploratory analyses also examined the relationship between MRI parameter change scores and clinical values, with results in eAppendix 3 and eFigures 1–3.

Data Availability

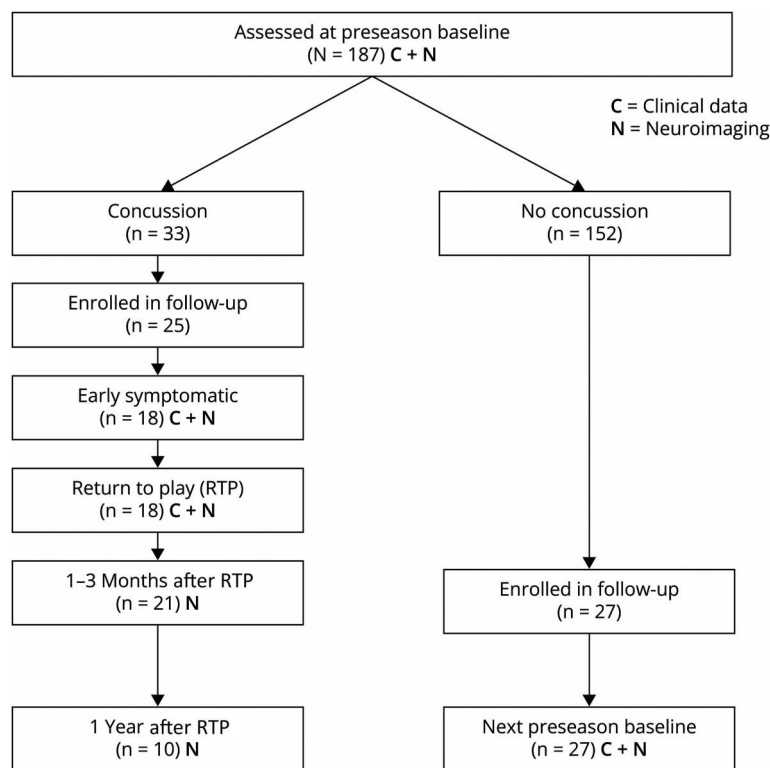
The authors have documented all data, methods, and materials used to conduct this research study, and anonymized data will be shared by request from any qualified investigator.

Results

Clinical and Demographic Data

Figure 1 shows the study workflow, with participant numbers per imaging session. A total of 187 athletes were enrolled in the study at baseline (mean age: 20.1 ± 2.0 years, 95/187 (51%) female and 92/187 (49%) male, 80/187 (43%) with HOC, 63/187 (34%) in collision sports); 33 were later concussed during the course of the study (mean age: 20.1 ± 2.3 years, 18/33 (55%) female and 15/33 (45%) male, 15/33 (45%) with HOC, 15/33 (45%) in collision sports), and of these, 25 were successfully imaged after their concussion. An

Figure 1 Study Design Flowchart



Time from baseline to follow-up: **BOS2:** 375 (364, 391) days
SYM: 70 (27, 197) days
RTP: 150 (82, 252) days
1-3 Months: 242 (137, 501) days

Plot illustrates recruitment design, assessment time points for concussed and control groups, and participant numbers at each assessment time point. Follow-up time intervals are summarized by median (lower quartile, upper quartile).

additional 27 uninjured athletes were reimaged as controls at their subsequent baseline. The Table shows that participant groups are from a similar age band, having comparable proportions of male and female athletes, with and without HOC, in collision and noncollision sports. The median time to full clinical recovery was approximately 3 weeks, with substantial interindividual variability. Comparing groups, there were no significant differences in demographics, concussion characteristics of the HOC subgroups, or baseline clinical scores. Examining attrition, there were no significant correlations between dropout at YR1 and demographic variables, nor acute symptom count or severity (all Spearman $|p| \leq 0.23$, $p \geq 0.268$).

Examining clinical data, athletes had low but nonzero scores for symptom severity and number of symptoms at baseline. At SYM, significantly elevated symptoms were seen (severity: $p = 0.001$, $z = 3.23$; number: $p < 0.001$, $z = 3.58$), but symptoms were no longer elevated at RTP (severity: $p = 0.999$, $z = -3.36$, number: $p > 0.999$, $z = -3.56$). For the other clinical measures, no significant declines in SAC scores or PS were observed at SYM or RTP (all $z \geq -0.60$, $p > 0.276$). In addition, no significant increases in balance errors, reaction time, or TMT A and B completion time were observed at an FDR threshold of 0.05, although at SYM, the M-BESS errors had increased at $p < 0.05$ unadjusted ($z = 1.79$, $p = 0.036$); for all other

measures and sessions, effects failed to reach even this threshold (all $z \leq 1.40$, $p \geq 0.081$). Thus, concussion effects were primarily symptomatic, without overt signs of impairment in balance or cognition. Furthermore, no evidence of post-RTP symptom reemergence was reported for concussed athletes by the sport medicine clinic.

CBF Data

For CBF (Figure 2, A–D), no significant changes were detected at SYM relative to baseline, but significant reductions were seen at RTP, mainly in bilateral insular, orbitofrontal, temporal, and parietal regions (mean difference and 95% CI $-8.97 [-12.80, -5.01]$ mL/100 g/minute, BSR = -4.53 , $p < 0.001$). At 1-3MO, the CBF declines were more spatially limited, being confined to the left parahippocampal gyrus ($-11.79 [-18.36, -4.34]$ mL/100 g/minute, BSR = -3.12 , $p = 0.002$), but at 1 YR, reductions were again seen in mainly insular and frontal regions ($-10.66 [-17.47, -5.69]$ mL/100 g/minute, BSR = -3.78 , $p < 0.001$). Figure 2E plots average CBF change scores of both controls and concussed athletes, for brain regions showing significant effects at RTP. Controls had a nonsignificant mean increase in these areas (BSR = 0.62 , $p = 0.524$). In comparison, concussed athletes had a mean decrease at all sessions, with effects that did not significantly differ from controls at SYM (BSR = -1.48 , $p = 0.130$), but did

Table Summary of Demographic and Clinical Data for Control and Concussed Participants

Demographic data	Controls	Concussed		z-score	p Value
Age (y)	19.7 ± 1.8	20.3 ± 1.5		−1.44	0.148
Sex (%)					
Female	15/27 (56)	11/25 (44)		0.42	0.416
Male	12/27 (44)	14/25 (56)			
HOC (%)	12/27 (44)	11/25 (44)		0.98	0.983
Collision sports (%)	11/27 (41)	14/25 (56)		0.28	0.281
Days to RTP	—	22 (12, 54)		—	—
HOC characteristics					
Number	2 (1, 2)	2 (1, 2)		0.03	0.973
Time since last (mo.)	14 (7, 28)	30 (21, 36)		−1.88	0.061
Recovery time (dy.)	31 (21, 41)	41 (15, 60)		−0.46	0.646
Clinical test data	Baseline	Baseline	ACU	RTP	
Symptom severity ^a	2 (1, 6)	5 (0, 14)	17 (6, 37) ^a	0 (0, 2)	−1.02 0.308
Number of symptoms ^a	2 (1, 4)	5 (0, 7)	9 (4, 14) ^a	0 (0, 2)	−1.13 0.255
SAC total score ^b	40 (35, 42)	39 (36, 43)	42 (38, 45)	43 (40, 46)	0.08 0.933
M-BESS total score	2 (0, 4)	3 (2, 6)	5 (1, 9)	1 (0, 5)	−1.29 0.194
SRT (ms) ^c	275 (260, 279)	260 (246, 274)	281 (250, 304)	258 (251, 278)	1.62 0.106
CRT (ms) ^c	375 (348, 413)	361 (339, 394)	359 (346, 396)	341 (326, 361)	0.69 0.488
PS ^c	75.5 (72.0, 86.0)	82.5 (67.0, 88.0)	75.0 (70.8, 82.0)	82.0 (78.0, 86.5)	−0.30 0.765
TMT-A (s) ^c	16.0 (12.2, 17.9)	16.8 (15.5, 20.4)	14.9 (12.9, 17.6)	12.3 (10.7, 14.2)	−1.54 0.125
TMT-B (s) ^c	31.3 (24.1, 34.6)	34.7 (27.0, 40.9)	30.4 (23.8, 33.9)	23.0 (18.7, 28.2)	−1.16 0.248

For participants with history of concussion (HOC), relevant characteristics are also reported. Age is summarized by mean ± SD, and clinical variables are summarized by median (lower quartile, upper quartile). Clinical data were collected at preinjury baseline, acute injury (ACU), and return to play (RTP). Total scores are reported for symptom total severity and number of symptoms, Standardized Assessment of Concussion (SAC), and Modified Balance Error Scoring System (M-BESS). For cognitive testing, reaction time points are reported for simple and choice reaction time (SRT and CRT), throughput scores are reported for processing speed (PS), and completion times are reported for Trail Making Tests A and B (TMT-A and TMT-B). The approximate z-scores and p values are also reported for group comparisons at baseline, based on 2-sample Wilcoxon tests.

^a Symptom scores are based on the 22-item list common to both SCAT3 and SCAT5.

^b SAC scores from the 21 of 27 controls and 17 of 25 concussed with SCAT5 data collected.

^c Cognitive scores from the 22 of 27 controls and 18 of 25 concussed with full C3 Logix data collected.

differ significantly at RTP (BSR = −3.40, $p = 0.002$). This was followed by diminished group differences at 1-3MO (−1.47, $p = 0.144$), with significant differences re-emerging at 1 YR (BSR = −2.79, $p = 0.010$). Whole-brain analysis of controls showed no significant longitudinal changes in CBF and no baseline differences relative to the concussed group.

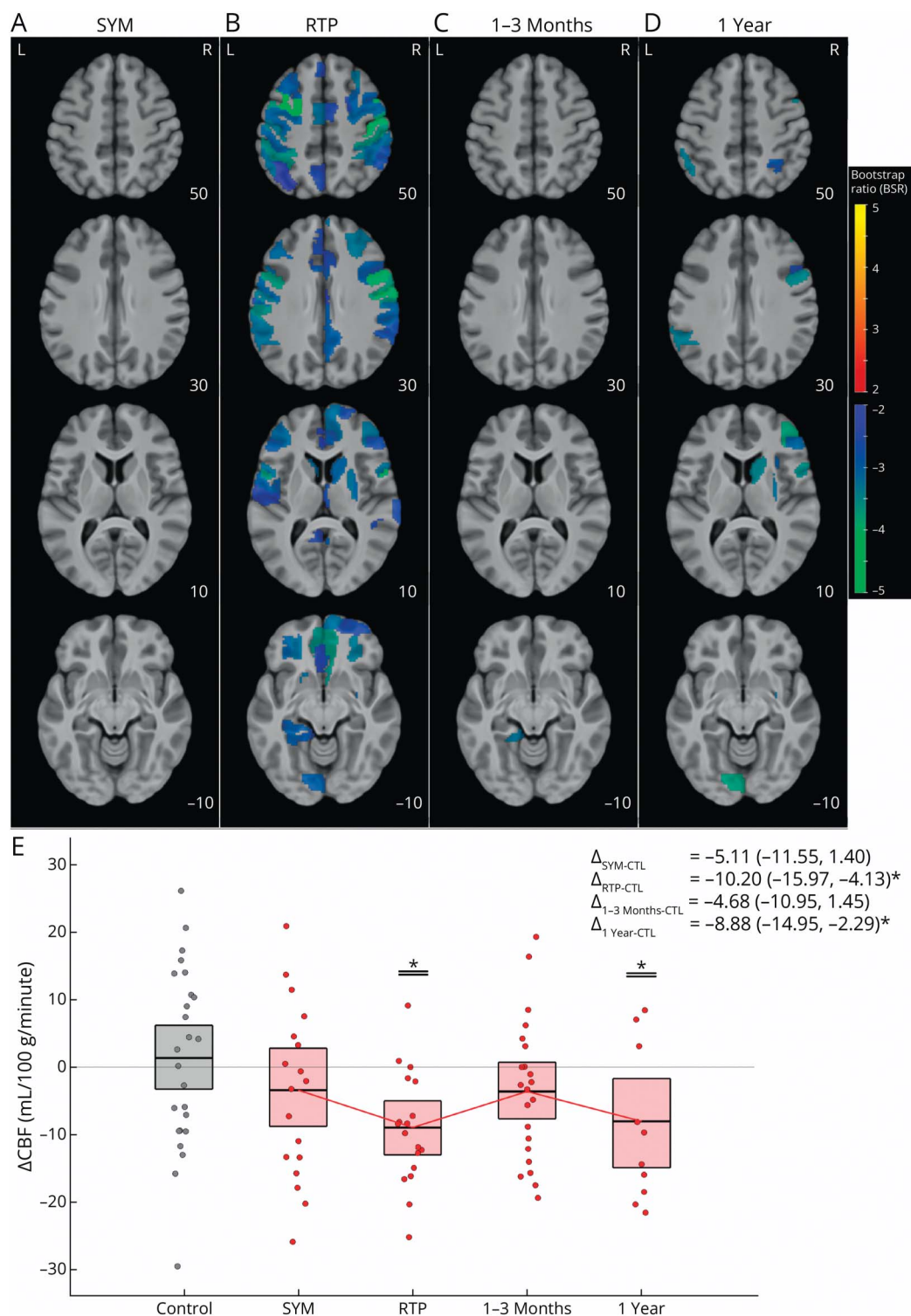
MD Data

For MD (Figure 3, A–D), spatially limited increases were seen at SYM (1.55×10^{-5} [0.81, 2.39] $\times 10^{-5}$, BSR = 3.86, $p < 0.001$) and more spatially extensive increases at RTP (1.94×10^{-5} [1.26, 2.69] $\times 10^{-5}$, BSR = 5.48, $p < 0.001$), with effects predominantly in the left corona radiata and internal capsule. The significant effects persisted at 1-3MO (1.71×10^{-5} [1.19, 2.29] $\times 10^{-5}$, BSR = 5.75, $p < 0.001$) and at 1 YR (2.12×10^{-5} [1.24, 3.03] $\times 10^{-5}$, BSR = 4.45, $p < 0.001$). Figure 3E

plots average MD change scores of both controls and concussed athletes, for brain regions showing significant effects at RTP. Controls had a significant mean increase in these areas (BSR = 2.37, $p = 0.024$). Concussed athletes had a greater mean increase at all sessions, but the effect did not significantly differ from controls at any postconcussion imaging sessions (all |BSRs| ≤ 1.18, $p \geq 0.238$). Whole-brain analysis of controls showed significant longitudinal increases in MD for 6 of 48 regions, including the fornix, corticospinal tract, superior cerebellar peduncle, posterior limb of the internal capsule, cingulum, and inferior fronto-occipital fasciculus, but no baseline differences were seen relative to the concussed group.

FA Data

For FA (Figure 4, A–D), effects were limited to the genu of the corpus callosum at SYM ($−5.64 \times 10^{-3}$ [−8.50, −2.46] $\times$

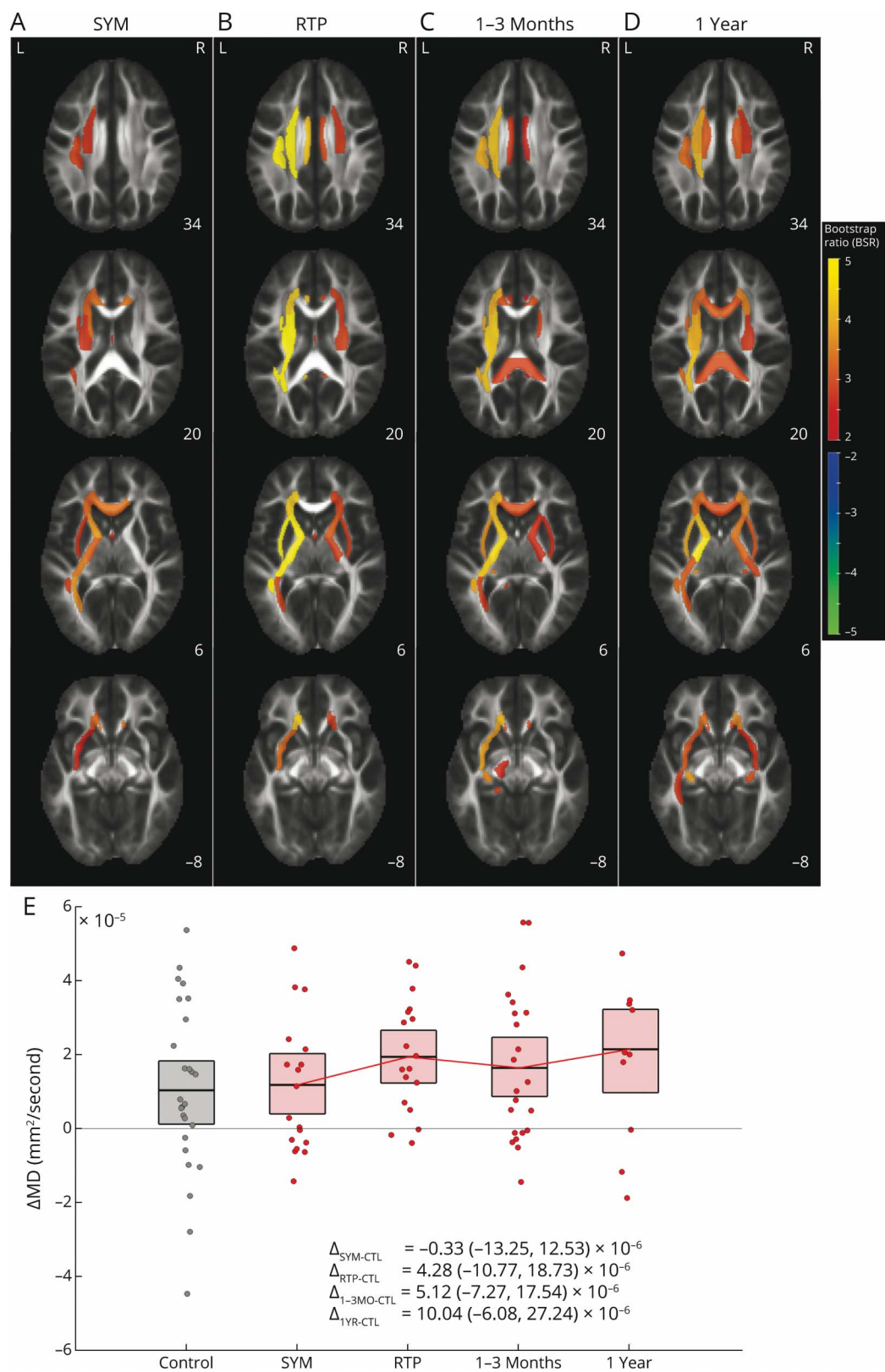
Figure 2 Regions of Significant Postconcussion Change in Cerebral Blood Flow (Δ CBF) Relative to Baseline

Results are shown at (A) SYM, (B) RTP, (C) 1-3MO, and (D) 1 YR. Axial slices are in MNI space, with the z-axis coordinates listed; for significant regions, standardized effect sizes are reported as z-distributed bootstrap ratio (BSR) values. For brain regions showing change at RTP, mean subject Δ CBF values are plotted (E). Horizontal solid black lines denote group means, and shaded boxes indicate 95% CIs of the mean. For concussed athletes, the distribution means are connected between sessions by solid red lines and zero effect is plotted as a dashed horizontal black line. MNI = Montreal Neurological Institute; RTP = return to play; SYM = symptomatic injury.

10^{-3} , BSR = -3.72, $p < 0.001$) but increased substantially in magnitude and spatial extent at RTP (-7.30×10^{-3} [-9.80, -5.05] $\times 10^{-3}$, BSR = -6.07, $p < 0.001$), again confined mainly

to the left corona radiata and internal capsule, mirroring the effects seen in MD. The significant effects remained at 1-3MO (-8.66×10^{-3} [-11.90, -5.70] $\times 10^{-3}$, BSR = -5.65, $p <$

Figure 3 Regions of Significant Postconcussion Change in Mean Diffusivity (Δ MD) Relative to Baseline



Results are shown at (A) SYM, (B) RTP, (C) 1-3MO, and (D) 1 YR. Axial slices are in MNI space, with the z-axis coordinates listed; for significant regions, standardized effect sizes are reported as z-distributed bootstrap ratio (BSR) values. For brain regions showing change at RTP, mean subject Δ MD values are plotted (E). Horizontal solid black lines denote group means, and shaded boxes indicate 95% CIs of the mean. For concussed athletes, the distribution means are connected between sessions by solid red lines and zero effect is plotted as a dashed horizontal black line. MNI = Montreal Neurological Institute; RTP = return to play; SYM = symptomatic injury.

0.001), but effects were no longer significant at 1 YR. Figure 4E plots average FA change scores of both controls and concussed athletes, for brain regions showing significant effects at RTP. Controls had a nonsignificant mean decrease in these areas ($BSR = -0.62, p = 0.566$). Concussed athletes had a greater mean decrease at all sessions, but the effect did not significantly differ from controls at any postconcussion imaging sessions (all $|BSRs| \leq 1.89, p \geq 0.060$). Whole-brain analysis of controls showed a significant longitudinal decrease in FA of the cerebellar peduncle, but no baseline differences were seen relative to the concussed group.

Analysis of Neuroimaging and Recovery Time

Analysis of correlations between time to RTP and neuroimaging measures only found significant effects for CBF, with results of these analyses depicted in Figure 5. Significant positive associations were identified within medial temporal areas: at SYM within the right fusiform gyrus ($A20rv$, center of mass (CoM) = 33, -15, -34; $\rho = 0.62$ [0.31, 0.83], $BSR = 4.82, p = 0.001$), at RTP in the left parahippocampal gyrus (TI , CoM = -23, 2, -32; $\rho = 0.64$ [0.44, 0.81], $BSR = 6.80, p < 0.001$), and at 1 YR within the left hippocampus ($rHipp$, CoM = -22, -14, -19; $\rho = 0.79$ [0.48, 0.99], $BSR = 5.17, p = 0.006$). Exploratory analyses in eAppendix 3 also found significant associations between MRI and clinical change scores, at SYM and RTP.

Discussion

There is ongoing debate about the extent that clinical indices reflect injury and recovery after a concussion^{1,2} and whether biological recovery is complete at medical clearance. This study expands on previous prospective imaging studies,^{9–12} by assessing CBF, MD, and FA of concussed athletes at preinjury baseline, early SYM, RTP, and up to 1 year afterward. Analyses show significant, spatially extensive changes in all MRI parameters at RTP relative to baseline, with effects that persist in later imaging sessions. Comparison with uninjured controls, however, shows that postconcussion changes only exceed normal longitudinal variability for CBF. Furthermore, changes in CBF alone show significant correlations with time to RTP.

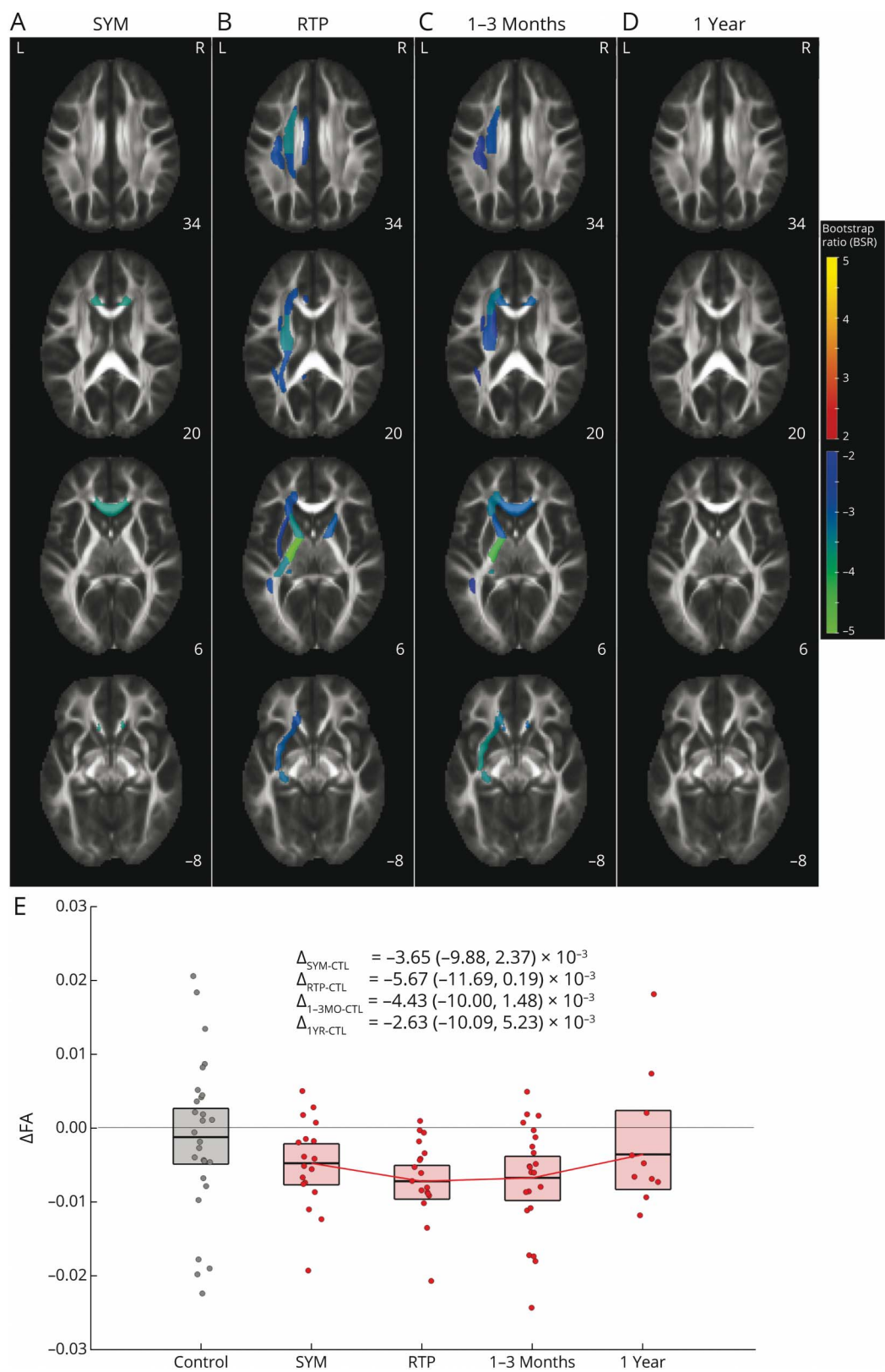
The ASL analyses found nonsignificant CBF effects at early injury, but spatially extensive changes at RTP, which diminished at 1–3 months after RTP before again increasing at 1 year after RTP. At early injury, previous cross-sectional studies have reported differing CBF responses, including significant decreases,²⁴ increases,²⁵ and nonsignificant differences.²⁶ Such initial alterations in CBF, due to disrupted autoregulation and microvascular injury, contribute to impaired neural function and subsequent biochemical cascades.¹³ The inconsistent effects may be due to sampling variability in days after injury,²⁶ given the rapidly evolving CBF response at early injury.¹³ For post-RTP effects, results may be compared to previous longitudinal studies with cross-sectional comparison.^{18,27} Both studies found limited CBF

effects in the first week of injury, as in this study, but also nonsignificant effects at RTP. For this study, the increased sensitivity at RTP may be due to the within-subject design. One study with imaging at 1 year after RTP¹⁸ also found significant declines in frontotemporal CBF, corroborating the results of this study. Prolonged decreases in CBF are of significant concern, likely reflecting ongoing secondary injury processes including neurometabolic impairment, neuroinflammation, and permeation of the blood-brain barrier.^{28,29} Such perfusion deficits can in turn exacerbate injuries, for example, through decreased protein synthesis and greater oxidative stress,^{29,30} contributing to long-term gray matter declines. Nevertheless, the greatest CBF reductions in this study coincided with RTP, despite a lack of overt behavioral decrements at this time, based on clinical testing.

The DTI analyses found significant increases in MD and decreases in FA, which increased in spatial extent and magnitude at RTP before gradually declining from 1 to 3 months to 1 year after RTP. Findings at early SYM are congruent with previous DTI literature in concussion and mild traumatic injury, which typically reports increased diffusivity and lower FA.^{26,31–33} These effects have been interpreted as signs of diffuse axonal injury³⁴ although other contributors may include cytotoxic and vasogenic edema.³⁵ The weaker early response may also be partly attributed to edema, which can cause variable changes in microstructure and resulting diffusion patterns.³⁶ Regarding post-RTP effects, results may again be compared with previous longitudinal studies with cross-sectional comparisons.^{32,37,38} In these studies, disturbances in FA and MD in the first week after injury were generally unresolved at follow-up, which ranged from 3 weeks to 6 months after injury, despite clinical resolution. In addition, a more recent cross-sectional analysis found increases in MD and decreases in FA from ACU to RTP, with MD effects further increasing at 1 year afterward but FA being no longer significant.¹⁸ These findings broadly align with the current prospective analyses. These persistent changes in DTI parameters are believed to reflect chronic injury-related processes including delayed inflammatory response³⁹ and axonal degeneration.⁴⁰ Despite these microstructural brain changes, all concussed athletes self-reported as asymptomatic at RTP and had returned to normal activities. This again suggests limited behavioral effects at this time, although the subjective nature of self-report may contribute to some underestimation of symptoms.

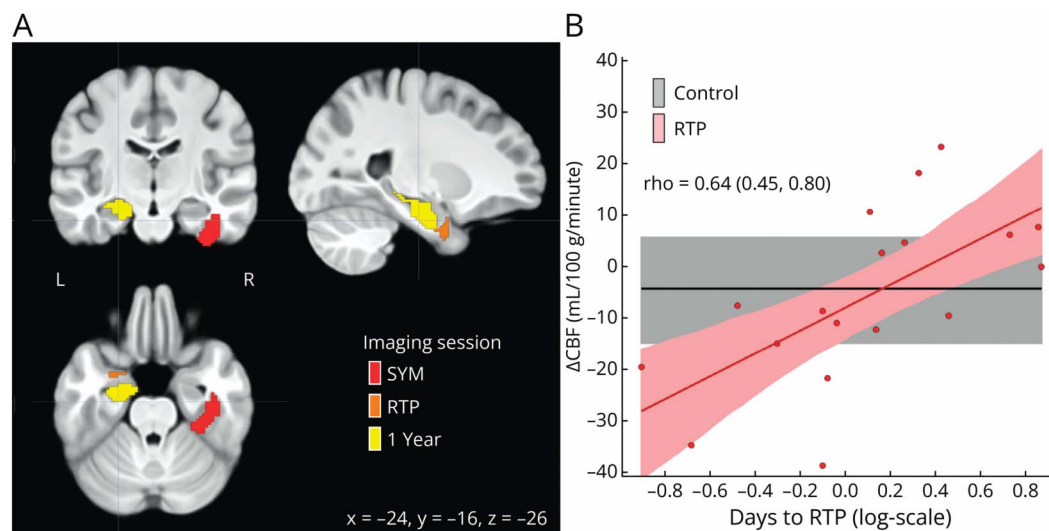
While analysis of CBF and white matter indices of MD and FA all showed significant longitudinal changes after RTP, the comparison of longitudinal brain changes in concussed athletes and controls showed that not all changes were concussion-specific. For CBF, uninjured controls showed modest but nonsignificant increases over time, consistent with previous literature¹⁹ and likely reflecting adaptation to exertion.⁴¹ Postconcussion changes proceeded in the opposite direction and differed significantly at RTP and 1 year after RTP, indicating a distinct physiologic response. For MD,

Figure 4 Regions of Significant Postconcussion Change in Fractional Anisotropy (Δ FA) Relative to Baseline



Results are shown at (A) SYM, (B) RTP, (C) 1-3MO, and (D) 1 YR. Axial slices are in MNI space, with the z-axis coordinates listed; for significant regions, standardized effect sizes are reported as z-distributed bootstrap ratio (BSR) values. For brain regions showing change at RTP, mean subject Δ FA values are plotted (E). Horizontal solid black lines denote group means, and shaded boxes indicate 95% CIs of the mean. For concussed athletes, the distribution means are connected between sessions by solid red lines and zero effect is plotted as a dashed horizontal black line. MNI = Montreal Neurological Institute; RTP = return to play; SYM = symptomatic injury.

Figure 5 Regions of Significant Correlation Between Postconcussion Change in Cerebral Blood Flow (Δ CBF) and Time to Medical Clearance



Results include significant effects at SYM, RTP, and 1 YR. For the brain maps (A), image slices are in MNI space, with x-axis, y-axis, and z-axis coordinates at the bottom of the panel. For significant regions, label colors denote the imaging session of significance. A representative scatterplot is shown for effects at RTP (B), where the red line represents a fitted regression line, and the shaded area indicates 95% CIs of the regression line. The black line represents the corresponding control mean, and the shaded area gives its 95% CI. MNI = Montreal Neurological Institute; RTP = return to play; SYM = symptomatic injury.

however, uninjured controls showed a significant increase, consistent with previous literature,¹⁹ and postconcussion changes were not significantly distinguishable from this group. These effects were mirrored in the FA analyses, where controls showed a nonsignificant decline but postconcussion changes were not significantly distinguishable from this group. These results indicate that although MD and FA are sensitive measures of postconcussion brain change, they may not differentiate from other sources of variability among athletes. This may include the cumulative effects of physical exertion, along with physiologic and psychological stress, which influence factors such as neuroinflammation and redox state⁴²⁻⁴⁴ that alter the microstructural environment.

Correlation analyses further examined the relationship between postconcussion changes in MRI measures and time to RTP, with only CBF showing significant associations. Previous cross-sectional analyses reported positive correlations between CBF and severity of the clinical outcome. This includes correlations between symptom severity and CBF in occipital, parietal, and medial temporal regions in early injury^{45,46}; correlations between time to medical clearance and temporal CBF at 1 year after RTP¹⁸; and enhanced global CBF among individuals with persistent postconcussion symptoms.⁴⁷ The current prospective analyses align with these findings, with positive associations between medial temporal CBF and time to medical clearance, seen both at early SYM and after RTP. This suggests that abnormally elevated perfusion in these regions may be a biomarker of more severe clinical outcome. This is potentially a consequence of abnormal blood flow autoregulation after injury, which in some cases enhances CBF,⁴⁸ although other mechanisms such

as neuroinflammation may contribute. Further work is needed to clarify the underlying mechanisms, for example, by measuring physiologic changes that precede CBF effects.

Although this study provided unprecedented insights into postconcussion recovery, some limitations should be acknowledged. There was longitudinal data loss, particularly at 1 year after RTP, which is relevant to assessing long-term recovery. Nevertheless, given large effect sizes and a lack of demographic or clinical correlations with attrition, results should be given careful consideration. Similarly, some clinical data were missing (neurocognitive tests) or could not be analyzed over the full sample (different version of the SAC); given uniformly small effect sizes for these tests, however, we may infer a lack of substantial cognitive deficits in the concussed group. For cross-sectional analyses, the determination of significant CBF and nonsignificant DTI effects depends on how normal brain variability was evaluated in the control group. This was obtained by comparing preseason baseline sessions with control sessions for within-season changes associated with exertional load.¹⁹ Nevertheless, this does not exactly reflect the trajectory of concussed athletes, in which season play is interrupted (along with school, work, and socialization), followed by gradual return to activities. The control group may also have unmodeled variability, for example, undisclosed concussions outside the sport setting. Future work should aim to prospectively recruit a more directly comparable cohort without brain injury, such as musculoskeletal injury, given similar interruptions to activities and overlapping symptom profiles.

Given the challenges of prospective study design, the sample size was reasonable, but further studies must replicate the

present findings. Larger samples are needed to probe heterogeneity in clinical presentation, for example, cognitive and symptom subdomains. These studies should also, where possible, use state-of-the-art imaging. For example, the pulsed ASL sequence in this study may be replaced with pseudo-continuous ASL⁴⁹ to enhance the sensitivity and reliability of CBF measurements. Other important considerations in future work include incorporating signs and symptoms proximal to the concussion event⁵⁰ and impact-monitoring devices⁵¹ to better identify concussion events and evaluate the contributing role of subconcussive blows. Finally, this work focused on sport-related concussion, where homogeneous demographics and injury biomechanics improve study power; this comes at the cost of being less representative of other cohorts and injuries. Hence, it is imperative that future prospective studies also focus on individuals outside the sporting context, to better understand how effects generalize to the broader population.^{52,53}

Conclusions

The multiparameter MRI data provide convergent evidence that physiologic brain recovery is incomplete at medical clearance, with significant ongoing alterations in perfusion and tissue microstructure. By controlling for baseline neurophysiology, this provides one of the most accurate characterizations of these brain changes to date. Findings in the absence of persistent clinical issues suggest that changes represent a functional “new normal”; it is not clear at present whether renormalization of brain physiology occurs on sufficiently long time lines, making this an important area of further research. Nevertheless, the presence of significant, long-lasting brain changes after injury reinforces concerns about the consequences of repeated concussions and to what extent these effects accumulate over time.

Author Contributions

N.W. Churchill: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M.G. Hutchison: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. S.J. Graham: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. T.A. Schweizer: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Disclosure

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References

1. Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport—Amsterdam, October 2022. *Br J Sports Med.* 2023;57(11):695-711. doi:10.1136/bjsports-2023-106898
2. Tabor JB, Brett BL, Nelson L, et al. Role of biomarkers and emerging technologies in defining and assessing neurobiological recovery after sport-related concussion: a systematic review. *Br J Sports Med.* 2023;57(12):789-797. doi:10.1136/bjsports-2022-106680
3. McKee AC, Cantu RC, Nowinski CJ, et al. Chronic traumatic encephalopathy in athletes: progressive tauopathy after repetitive head injury. *J Neuropathol Exp Neurol.* 2009;68(7):709-735. doi:10.1097/NEN.0b013e3181a9d503
4. Smith DH, Johnson VE, Stewart W. Chronic neuropathologies of single and repetitive TBI: substrates of dementia? *Nat Rev Neurol.* 2013;9(4):211-221. doi:10.1038/nrneurol.2013.29
5. Kodiweera C, Alexander AL, Harezlak J, McAllister TW, Wu Y-C. Age effects and sex differences in human brain white matter of young to middle-aged adults: a DTI, NODDI, and q-space study. *Neuroimage.* 2016;128:180-192. doi:10.1016/j.neuroimage.2015.12.033
6. Churchill N, Hutchison M, Richards D, Leung G, Graham S, Schweizer TA. Brain structure and function associated with a history of sport concussion: a multi-modal magnetic resonance imaging study. *J Neurotrauma.* 2017;34(4):765-771. doi:10.1089/neu.2016.4531
7. Liu Y, Zhu X, Feinberg D, et al. Arterial spin labeling MRI study of age and gender effects on brain perfusion hemodynamics. *Magn Reson Med.* 2012;68(3):912-22. doi:10.1002/mrm.23286
8. Chauvin L, Kumar K, Wachinger C, et al. Neuroimage signature from salient key-points is highly specific to individuals and shared by close relatives. *Neuroimage.* 2020;204:116208. doi:10.1016/j.neuroimage.2019.116208
9. Pasternak O, Koerte IK, Bouix S, et al. Hockey Concussion Education Project, Part 2. Microstructural white matter alterations in acutely concussed ice hockey players: a longitudinal free-water MRI study. *J Neurosurg.* 2014;120(4):873-881. doi:10.3171/2013.12.JNS132090
10. Chamard E, Théoret H, Skopelja EN, Forwell LA, Johnson AM, Echlin PS. A prospective study of physician-observed concussion during a varsity university hockey season: metabolic changes in ice hockey players. Part 4 of 4. *Neurosurg Focus.* 2012;33(6):1-7. doi:10.3171/2012.10.FOCUS12305
11. Jarrett M, Tam R, Hernández-Torres E, et al. A prospective pilot investigation of brain volume, white matter hyperintensities, and hemorrhagic lesions after mild traumatic brain injury. *Front Neurol.* 2016;7:11. doi:10.3389/fneur.2016.00011
12. Koerte IK, Kaufmann D, Hartl E, et al. A prospective study of physician-observed concussion during a varsity university hockey season: white matter integrity in ice hockey players. Part 3 of 4. *Neurosurg Focus.* 2012;33(6):1-7. doi:10.3171/2012.10.FOCUS12303
13. Giza CC, Hovda DA. The new neurometabolic cascade of concussion. *Neurosurgery.* 2014;75 Suppl 4(0 4):S24-S33. doi:10.1227/NEU.0000000000000505
14. Sullivan DR. A cerebrovascular hypothesis of neurodegeneration in mTBI. *J Head Trauma Rehabil.* 2019;34(3):E18-E27. doi:10.1097/HTR.0000000000000449
15. Ng SY, Lee AYW. Traumatic brain injuries: pathophysiology and potential therapeutic targets. *Front Cell Neurosci.* 2019;13:528. doi:10.3389/fncel.2019.00528
16. Tayebi M, Holdsworth SJ, Champagne AA, et al. The role of diffusion tensor imaging in characterizing injury patterns on athletes with concussion and subconcussive injury: a systematic review. *Brain Inj.* 2021;35(6):621-644. doi:10.1080/02699052.2021.1895313
17. Wang Y, Bartels HM, Nelson LD. A systematic review of ASL perfusion MRI in mild TBI. *Neuropsychol Rev.* 2023;33(1):160-191. doi:10.1007/s11065-020-09451-7
18. Churchill NW, Hutchison MG, Graham SJ, Schweizer TA. Mapping brain recovery after concussion: from acute injury to 1 year after medical clearance. *Neurology.* 2019;93(21):e1980-e1992. doi:10.1212/WNL.00000000000008523
19. Churchill NW, Hutchison MG, Graham SJ, Schweizer TA. Disturbances in brain physiology due to season play: a multi-sport study of male and female university athletes. *Front Physiol.* 2021;12:653603. doi:10.3389/fphys.2021.653603
20. equator-network.org/reporting-guidelines/strobe
21. Brainnetome Atlas. Accessed February 28, 2025. atlas.brainnetome.org/
22. Johns Hopkins University ICBM-DTI-81 white matter atlas. Accessed February 28, 2025. cmrm.med.jhmi.edu/
23. Simon M, Maerlender A, Metzger K, Decoster L, Hollingworth A, Valovich McLeod T. Reliability and concurrent validity of select C3 logit test components. *Dev Neuropsychol.* 2017;42(7-8):446-459. doi:10.1080/87565641.2017.1383994
24. Wang Y, Nencka AS, Meier TB, et al. Cerebral blood flow in acute concussion: preliminary ASL findings from the NCAA-DoD CARE consortium. *Brain Imaging Behav.* 2019;13(5):1375-1385. doi:10.1007/s11682-018-9946-5
25. Churchill NW, Hutchison MG, Graham SJ, Schweizer TA. Mapping brain recovery after concussion: from acute injury to 1 year after medical clearance. *Neurology.* 2019;93(21):e1980-e1992. doi:10.1212/WNL.00000000000008523

26. Churchill NW, Hutchison MG, Richards D, Leung G, Graham SJ, Schweizer TA. The first week after concussion: blood flow, brain function and white matter microstructure. *Neuroimage Clin.* 2017;14:480-489. doi:10.1016/j.nicl.2017.02.015
27. Meier TB, Bellgowan PS, Singh R, Kuplicki R, Polanski DW, Mayer AR. Recovery of cerebral blood flow following sports-related concussion. *JAMA Neurol.* 2015;72(5):530-538. doi:10.1001/jamaneurol.2014.4778
28. Logsdon AF, Lucke-Wold BP, Turner RC, Huber JD, Rosen CL, Simpkins JW. Role of microvascular disruption in brain damage from traumatic brain injury. *Compr Physiol.* 2015;5(3):1147-1160. doi:10.1002/cphy.c140057
29. Sandsmark DK, Bashir A, Wellington CL, Diaz-Arrastia R. Cerebral microvascular injury: a potentially treatable endophenotype of traumatic brain injury-induced neurodegeneration. *Neuron.* 2019;103(3):367-379. doi:10.1016/j.neuron.2019.06.002
30. Truettner JS, Alonso OF, Dietrich WD. Influence of therapeutic hypothermia on matrix metalloproteinase activity after traumatic brain injury in rats. *J Cereb Blood Flow Metab.* 2005;25(11):1505-1516. doi:10.1038/sj.jcbfm.9600150
31. Murugavel M, Cubon V, Putukian M, et al. A longitudinal diffusion tensor imaging study assessing white matter fiber tracts after sports-related concussion. *J Neurotrauma.* 2014;31(22):1860-1871. doi:10.1089/neu.2014.3368
32. Churchill NW, Hutchison MG, Richards D, Leung G, Graham SJ, Schweizer TA. Neuroimaging of sport concussion: persistent alterations in brain structure and function at medical clearance. *Scientific Rep.* 2017;7(1):8297. doi:10.1038/s41598-017-07742-3
33. Inglese M, Makani S, Johnson G, et al. Diffuse axonal injury in mild traumatic brain injury: a diffusion tensor imaging study. *J Neurosurg.* 2005;103(2):298-303. doi:10.3171/jns.2005.103.2.0298
34. Povlishock JT, Christman CW. The pathobiology of traumatically induced axonal injury in animals and humans: a review of current thoughts. *J Neurotrauma.* 1995;12(4):555-564. doi:10.1089/neu.1995.12.555
35. Barzó P, Marmarou A, Fatouros P, Hayasaki K, Corwin F. Contribution of vasogenic and cellular edema to traumatic brain swelling measured by diffusion-weighted imaging. *J Neurosurg.* 1997;87(6):900-907. doi:10.3171/jns.1997.87.6.0900
36. Wilde E, McCauley S, Hunter J, et al. Diffusion tensor imaging of acute mild traumatic brain injury in adolescents. *Neurology.* 2008;70(12):948-955. doi:10.1212/01.wnl.0000305961.68029.54
37. Meier TB, Bergamino M, Bellgowan PS, et al. Longitudinal assessment of white matter abnormalities following sports-related concussion. *Hum Brain Mapp.* 2016;37(2):833-845. doi:10.1002/hbm.23072
38. Henry LC, Tremblay J, Tremblay S, et al. Acute and chronic changes in diffusivity measures after sports concussion. *J Neurotrauma.* 2011;28(10):2049-2059. doi:10.1089/neu.2011.1836
39. Streit WJ, Mrak RE, Griffin WST. Microglia and neuroinflammation: a pathological perspective. *J Neuroinflamm.* 2004;1:14. doi:10.1186/1742-2094-1-14
40. Song S-K, Sun S-W, Ju W-K, Lin S-J, Cross AH, Neufeld AH. Diffusion tensor imaging detects and differentiates axon and myelin degeneration in mouse optic nerve after retinal ischemia. *Neuroimage.* 2003;20(3):1714-1722. doi:10.1016/j.neuroimage.2003.07.005
41. Ainslie PN, Cotter JD, George KP, et al. Elevation in cerebral blood flow velocity with aerobic fitness throughout healthy human ageing. *J Physiol.* 2008;586(16):4005-4010. doi:10.1113/jphysiol.2008.158279
42. Munhoz C, Garcia-Bueno B, Madrigal J, Lepesch L, Scavone C, Leza J. Stress-induced neuroinflammation: mechanisms and new pharmacological targets. *Braz J Med Biol Res.* 2008;41(12):1037-1046. doi:10.1590/s0100-879x2008001200001
43. Bailey DM, Evans KA, McEneny J, et al. Exercise-induced oxidative-nitrosative stress is associated with impaired dynamic cerebral autoregulation and blood-brain barrier leakage. *Exp Physiol.* 2011;96(11):1196-1207. doi:10.1113/expphysiol.2011.060178
44. Jahangiri Z, Gholamnezhad Z, Hosseini M, Beheshti F, Kasraie N. The effects of moderate exercise and overtraining on learning and memory, hippocampal inflammatory cytokine levels, and brain oxidative stress markers in rats. *J Physiol Sci.* 2019;69(6):993-1004. doi:10.1007/s12576-019-00719-z
45. Churchill NW, Hutchison MG, Graham SJ, Schweizer TA. Symptom correlates of cerebral blood flow following acute concussion. *Neuroimage Clin.* 2017;16:234-239. doi:10.1016/j.nicl.2017.07.019
46. Lin C-M, Tseng Y-C, Hsu H-L, et al. Arterial spin labeling perfusion study in the patients with subacute mild traumatic brain injury. *PLoS One.* 2016;11(2):e0149109. doi:10.1371/journal.pone.0149109
47. Barlow KM, Marcil LD, Dewey D, et al. Cerebral perfusion changes in post-concussion syndrome: a prospective controlled cohort study. *J Neurotrauma.* 2017;34(5):996-1004. doi:10.1089/neu.2016.4634
48. Len T, Neary J. Cerebrovascular pathophysiology following mild traumatic brain injury. *Clin Physiol Funct Imaging.* 2011;31(2):85-93. doi:10.1111/j.1475-097X.2010.00990.x
49. Alsop DC, Detre JA, Golay X, et al. Recommended implementation of arterial spin-labeled perfusion MRI for clinical applications: a consensus of the ISMRM perfusion study group and the European consortium for ASL in dementia. *Magn Reson Med.* 2015;73(1):102-116. doi:10.1002/mrm.25197
50. Echemendia RJ, Ahmed OH, Bailey CM, et al. Introducing the Concussion Recognition Tool 6 (Crt6). *Br J Sports Med.* 2023;57(11):689-691. doi:10.1136/bjsports-2023-106851
51. Patton DA, Huber CM, Jain D, et al. Head impact sensor studies in sports: a systematic review of exposure confirmation methods. *Ann Biomed Eng.* 2020;48(11):2497-2507. doi:10.1007/s10439-020-02642-6
52. Woodrow RE, Winzeck S, Luppi AI, et al. Acute thalamic connectivity precedes chronic post-concussive symptoms in mild traumatic brain injury. *Brain.* 2023;146(8):3484-3499. doi:10.1093/brain/awad056
53. Richter S, Winzeck S, Correia MM, et al. Predicting recovery in patients with mild traumatic brain injury and a normal CT using serum biomarkers and diffusion tensor imaging (CENTER-TBI): an observational cohort study. *EClinicalMedicine.* 2024;75:102751. doi:10.1016/j.eclinm.2024.102751